

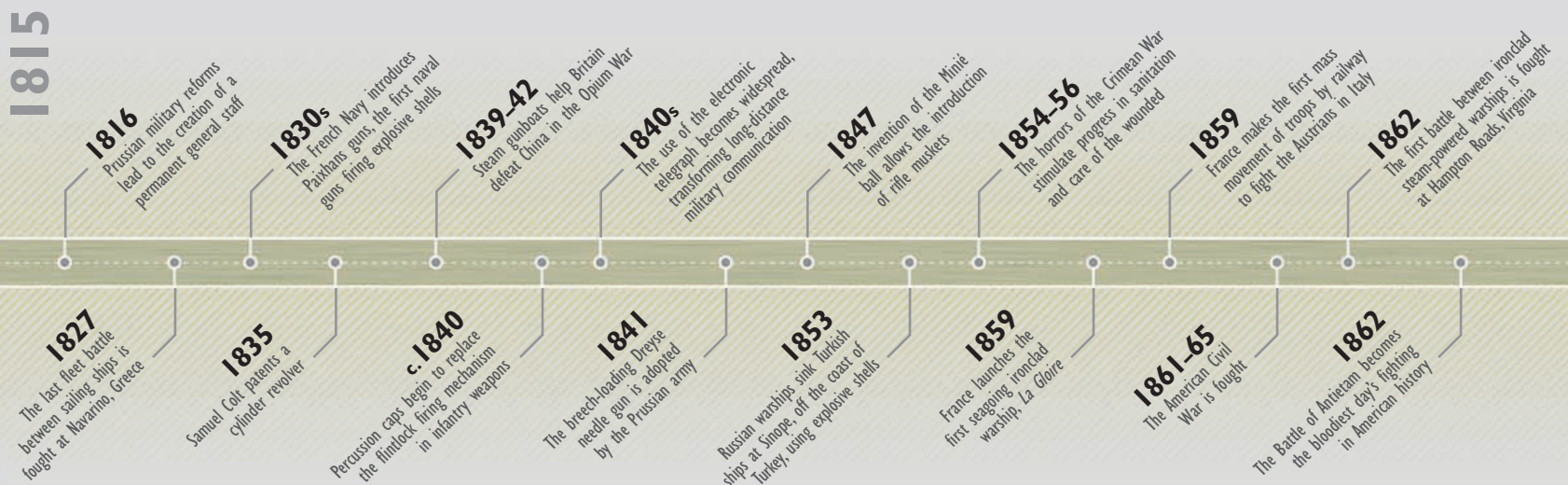
INTRODUCTION

Between 1815 and 1914, technological progress transformed conflict. High-explosive and rapid-fire mechanisms greatly increased the killing power of weaponry; railways enabled greater mobility; the telegraph, telephone, and radio revolutionized communications; and warfare spread into the skies and beneath the oceans.

Developments were relatively slow at first, but, in the 1830s, a forty-year transition period began, in which many hybrid and experimental technologies flourished. The long reign of the flintlock musket came to an end, though at first, mostly replaced by the rifle musket, which was still muzzle-loaded and slow-firing. Breech-loading rifles and pistols capable of repeated fire were invented, but armies were slow to accept them. At sea, ships used both steam engines and sails, and their wooden hulls were armoured with iron cladding. Even in the American Civil War of 1861–65, more soldiers still died of disease than in combat – as in wars of old.

The pace of change increased from the 1870s. The Franco-Prussian War of 1870–71 was fought by infantry with breech-loading rifles; the French fielded machine guns, while German rifled, breech-loading Krupp cannon bombarded Paris. In the 1880s, the use of smokeless propellants and high explosives in shells marked the end of the gunpowder era. New navies were built, with steel ships equipped with breech-loading rifled turret guns, as bolt-action repeater rifles became standard infantry equipment. Army organization was also changing, with the widespread adoption of the Prussian model of a professional general staff, and European powers committing to military service as a basic duty of their male citizens. More advanced nations amassed vast army and navy establishments, using them with success against tribal peoples and technologically inferior states, such as China or Turkey. The Russo-Japanese War of 1904–05 provided a glimpse of 20th-century conflict between industrialized states, with the presence of machine guns and heavy artillery, barbed wire, and trenches.

KEY DATES



THE AMERICAN CIVIL WAR – 1861–1865



THE FRANCO-PRUSSIAN WAR – 1870–71



THE BATTLE OF MANILA – 1898

